

FORMULAE

Module - 1 to 5

Module - 1

Eigenvalue & Eigenvector:

$$|A - \lambda I| = 0$$

characteristic Eqn

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0.$$

S_1 = Sum of main diagonal elements.

S_2 = Sum of minors of main-diagonals.

S_3 = $|A|$ = det |given Matrix|.

For eigenvector: $(A - \lambda I) X = 0.$

Diagonalization:

$$\text{Matrix } \Rightarrow D = P^{-1} A P$$

Module - 2

Fourier Series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

for $\boxed{0 < x < 2l}$ (or) $-l < x < l$.

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx; \quad a_n = \frac{1}{l} \int_0^{2l} f(x) \cdot \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \cdot \sin \frac{n\pi x}{l} dx.$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

for $0 < x < 2\pi$ (or) $-\pi < x < \pi$.

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx; \quad a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cdot \cos nx dx; \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cdot \sin nx dx.$$

bernoulli's : ~~u~~ ~~for~~

$$\int u v dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

odd/even check : $f(-x) = -f(x) \Rightarrow$ odd function
 $f(-x) = f(x) \Rightarrow$ even function.

$\frac{\text{odd}}{\text{odd}} = \text{even fn.}$		$\frac{\text{odd}}{\text{even}} = \text{odd fn.}$		$e^{\infty} = \infty,$
$\frac{\text{even}}{\text{even}} = \text{even fn.}$		$\frac{\text{even}}{\text{odd}} = \text{odd. fn.}$		$e^{-\infty} = 0$

odd $\times$ even = odd		even $\times$ odd = odd.
odd $\times$ odd = even		even $\times$ even = even.

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \cos ax dx = \frac{\sin ax}{a}$$

$$\int \sin ax dx = -\frac{\cos ax}{a}$$

Module - 3

$\Rightarrow$ Fourier Transform:

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cdot e^{+isx} dx.$$

Inverse Fourier Transform:

$$\begin{aligned} e^{i\theta} &= \cos\theta - i\sin\theta \\ e^{-i\theta} &= \cos\theta + i\sin\theta \end{aligned}$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) \cdot e^{-isx} ds$$

Parseval's identity :

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$$

Module - 4

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) \cdot z^{-n}$$

$$\Rightarrow 1 + x + x^2 + x^3 + \dots = (1-x)^{-1} \text{ if } |x| < 1$$

$$Z[a] = a \cdot \frac{z}{z-1} \quad \text{or} \quad Z[1] = \frac{z}{z-1}$$

Convolution theorem: $f(n) * g(n) = \sum_{r=0}^n f(r) g(n-r)$

Formulas used;

$$Z^{-1}\left(\frac{z}{z-a}\right) = a^n; \quad Z^{-1}\left(\frac{z}{z+a}\right) = (-a)^n$$

$$Z^{-1}\left[\frac{z}{(z-a)^2}\right] = na^{n-1}$$

+ Partial Fractions

Module - 5

$$Z[y_n] = F(z)$$

$$Z[y_{n+1}] = F(z) - y_0$$

$$Z[y_{n+2}] = F(z) - y_0 - \frac{y_1}{z}$$

$$Z[y_{n+3}] = F(z) - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2}$$

Differential And Difference Equations

Formulas:

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

← Important

S_1 = sum of diagonal elements

S_2 = sum of ^{Minors} ~~diagonals~~ of principal diagonal elements

$$S_3 = |A|$$

$$\cos n\pi = (-1)^n$$

$$\sin n\pi = 0$$

Module - 1

Q. $A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$: Find Eigenvalues & eigenvectors

Sol

Step-1: Find eigen values:

$$\Rightarrow |A - \lambda I| = 0$$

Step-2: Find characteristic equation:-

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

S_1 = sum of diagonal elements

S_2 = sum of minors of diagonal elements

$$S_3 = |A|$$

$$\begin{bmatrix} 2-\lambda & 0 & 1 \\ -1 & 4-\lambda & -1 \\ -1 & 2 & -\lambda \end{bmatrix}$$

$$S_1 = 2 + 4 + 0 \quad S_2 = (+2) + (+1) + (-0)$$

$$S_1 = 6$$

$$S_2 = 11$$

$$S_3 = |A| = (2(0+2) - 0 + 1(-2+4))$$

$$= 4 - 0 + 2 = 6 = S_3$$

Characteristic Equation is given by:

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

Step-3: Solve: $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$$\lambda = 1, 2, 3$$

$$\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$$

Step-4: Finding eigenvectors

For $\lambda = 1$,

$$\begin{bmatrix} 2-1 & 0 & 1 \\ -1 & 4-1 & -1 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ -1 & 3 & -1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 + x_3 = 0$
 $x_1 = -x_3$
 $x_2 = 0$

For $\lambda = 1$, Eigen vector = $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Step-5: $\lambda = 2$ = $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

Step-6: $\lambda = 3$ = $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

Hence, the eigen values are:
= 1, 2, 3

and the eigen vectors are:

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Practice: Find eigen values & eigen vectors:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Solution:

Characteristic Eqn.

$$(A - \lambda I) = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{bmatrix} = 0$$

S_1 = sum of diagonals

S_2 = sum of minors of diagonals

S_3 = $|A|$

$$S_1 = 2 + 2 + 2 = 6$$

$$S_2 = 4 + 4 + 4 = 12$$

$$S_3 = 2(4) - 1(0) + 0 = 8$$

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

Solving - we get;

$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 2$$

$$\lambda = 2, 2, 2$$

$$\lambda = 2, 2, 2$$

Finding eigenvectors:

For $\lambda = 2$

$$\begin{bmatrix} 2-2 & 1 & 0 \\ 0 & 2-2 & 1 \\ 0 & 0 & 2-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = 0$$

$$x_3 = 0$$

$$0 \cdot x_1 + x_2 + 0 \cdot x_3 = 0$$

$$0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 0$$

$$\frac{x_1}{1-0} = \frac{x_2}{0-0} = \frac{x_3}{0-0}$$

Learn essential properties



Check this out.

⇒ Cayley Hamilton Theorem: $A = \lambda$

If $A^n + c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + c_3 \lambda^{n-3} + \dots + c_n = 0$ is the char. eqn of a square matrix A of order n , then according to Cayley Hamilton Theorem, $A^n + c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_n I = 0$.

∴ Put $A = \lambda$

Q- Verify Cayley Hamilton Theorem for $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.
Also, find A^{-1} using it.

~~$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$ where $s_i = \text{sum of } i^{\text{th}} \text{ principal}$~~

Sol

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0.$$

$$s_1 = 1+2+1 = \boxed{4}; s_2 = -20; s_3 = 35$$

Characteristic Eqn:

$$\lambda^3 - 4\lambda^2 - 20\lambda + 35 = 0$$

According to Cayley Hamilton theorem: Put $\lambda = A$

$$A^3 - 4A^2 - 20A + 35I = 0 \quad \text{--- (1)}$$

Now, find

A^2 and A^3

$$A^2 = A \cdot A = \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix}$$

$$A^3 = A \cdot A^2 = \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix}$$

Now, substitute the values of A^2 and A^3 in Eq (1):

$$\Rightarrow A^3 - 4A^2 - 20A + 35I$$

$$\Rightarrow \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix} - 4 \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix} - 20 \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} + 35 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0.$$

⇒ ~~135~~ ... If the LHS and RHS = 0 equals to zero then, Cayley-Hamilton Theorem is verified.

Diagonalisation of a Matrix by means of orthogonal transformation.

Q $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$; Diagonalise the matrix & hence find A^4 .

Sol Step-1: Characteristic Eqn: $|A - \lambda I| = 0$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0.$$

Char. Eqn:

$$\lambda^3 - 18\lambda^2 + 45\lambda - 0 = 0.$$

$$\lambda = 15, 0, 3$$

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$$

$$s_1 = 8+7+3 = \boxed{18}$$

$$s_2 = \cancel{2+16+5} = \boxed{45}$$

$$s_3 = |A| = 8(21-16) + 6(-18+8) + 2(40) = 8(5) + 6(-10) + 2(40) = 40 - 60 + 80 = \boxed{60}$$

$$= 8(5) + 6(-10) + 2(40) = 40 - 60 + 80 = \boxed{60}$$

$$= \boxed{0}$$

Step-2: Now, find Eigen vectors:

Case-I: $(A - \lambda I)X = 0$.

For $\lambda = 0$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} (8-\lambda)x_1 - 6x_2 + 2x_3 = 0 \\ -6x_1 + (7-\lambda)x_2 - 4x_3 = 0 \\ 2x_1 - 4x_2 + (3-\lambda)x_3 = 0 \end{cases}$$

$$8x_1 - 6x_2 + 2x_3 = 0 \quad \text{--- (1)}$$

$$-6x_1 + 7x_2 - 4x_3 = 0 \quad \text{--- (2)}$$

$$-2x_1 - 4x_2 + 3x_3 = 0 \quad \text{--- (3)}$$

Solve Case

$$x_1 = \begin{bmatrix} 10 \\ 20 \\ 20 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Similarly

Case-II: $\lambda = 3$

$$\text{Eigenvector } x_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

Case-III: $\lambda = 15$

$$\text{Eigenvector } x_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix}$$

$$\begin{cases} (8-\lambda)x_1 - 6x_2 + 2x_3 = 0 \\ -6x_1 + (7-\lambda)x_2 - 4x_3 = 0 \\ 2x_1 - 4x_2 + (3-\lambda)x_3 = 0 \end{cases}$$

Mathematics Kala

YT

$$\frac{x_1}{24-14} = \frac{x_2}{-12+32} = \frac{x_3}{56-36}$$

$$\frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20}$$

Check diagonalization by orthogonal reduction, if the λ values are different then they are mutually orthogonal.

Step-3: Here, $x_1 = \begin{bmatrix} 10 \\ 20 \\ 20 \end{bmatrix}$ $x_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ $x_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

Here, all eigenvectors are orthogonal to each other.

Step-3: Form normalized modal matrix (N).

$$N = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$$

$$D = N^T A N \text{ or } \lambda = P^T A P$$

$$x_1 = \begin{bmatrix} 10 \\ 20 \\ 20 \end{bmatrix} \xrightarrow{1/3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \xrightarrow{1/3}$$

$$N = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{-2}{3} & \frac{1}{3} \end{bmatrix}$$

$$= \frac{1}{\sqrt{4+4+1}} = \frac{1}{3}$$

$$x_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$= \frac{2}{\sqrt{4+1+4}} = \frac{2}{3}$$

So $P^T A P$

$$= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{-2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{-2}{3} & \frac{1}{3} \end{bmatrix}$$

The resultant, we will get the diagonals as the eigen values.

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

To find A^4 :

we know that $D = P^T A P$

$$A = P D P^T$$

$$(A)^n = P (D)^n P^T$$

$$A^4 = P D^4 P^T$$

$$= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & (3)^4 & 0 \\ 0 & 0 & (15)^4 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 22536 & -22482 & 11214 \\ -22482 & 22509 & -11268 \\ 11214 & -11268 & 5661 \end{bmatrix}$$

$\therefore$ If eigen values are different then they have orthogonal v/c's.

Fourier Series:-

$$0 < x < 2l \quad \text{or} \quad -l < x < l$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{l}x\right)$$

for numbers or constants

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx \quad \left| \quad a_n = \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi}{l}x\right) dx \right.$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

period = $2\pi = 2l$

period = upper limit - lower limit

$$0 < x < 2\pi \quad \text{or} \quad -\pi < x < \pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx \quad \left| \quad a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \right.$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Bernoulli's

$$\begin{matrix} -a < x < a \\ -\pi < x < \pi \end{matrix}$$

$$f(-x) = f(x) \rightarrow \text{Even function}$$

$$f(-x) = -f(x) \rightarrow \text{odd function}$$

cos functions

$$\int u v dx = u v_1 - u v_2 + u v_3 - \dots$$

$$\frac{d}{dx} (x^n) = n x^{n-1}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \cos ax dx = \frac{\sin ax}{a}$$

$$\int \sin ax dx = -\frac{\cos ax}{a}$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

Q. Find Fourier series of $f(x) = x(2l - x)$ in $(0, 2l)$.
Deduce the sum of $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$

Sol $0 < x < 2l$

Formula: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$

$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$
 $a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$
 $b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$

$a_0 = \frac{1}{l} \int_0^{2l} x(2l - x) dx$

Bernoulli
 $\int u v dx = u v_1 - u' v_2 + u'' v_3$

$\int_0^{2l} (2xl - x^2) dx = \left[\frac{2x^2 l}{2} - \frac{x^3}{3} \right]_0^{2l}$

$= \frac{2(2l)^2 l}{2} - \frac{(2l)^3}{3}$

$= \frac{48l^3}{2} - \frac{8l^3}{3}$

$\frac{12l^3 - 8l^3}{3}$

$= \frac{1}{l} \left(\frac{4l^3}{3} \right)$

$\frac{4l^2}{3}$

~~$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$~~

$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$

$f(x) = 2xl - x^2$

~~$= \frac{1}{l} \int_0^{2l} (2xl - x^2) \cos \frac{n\pi x}{l} dx$~~

$a_n = \frac{1}{l} \int_0^{2l} (2xl - x^2) \cos \frac{n\pi x}{l} dx$

~~$a_n = \frac{1}{l} \int_0^{2l} (2xl - x^2) \cos \frac{n\pi x}{l} dx$~~

$a_n = \frac{1}{l} \left[(2xl - x^2) \left(\frac{\sin n\pi x}{\frac{n\pi}{l}} \right) - (2l - 2x) \left(\frac{-\cos n\pi x}{\frac{n^2 \pi^2}{l^2}} \right) \right. \right.$

$\left. + (-2) \left(\frac{-\sin n\pi x}{\frac{n^3 \pi^3}{l^3}} \right) \right]_0^{2l}$

$a_n = \frac{1}{l} \left[0 - (2l - 4l) \left(\frac{-\cos n\pi l}{\frac{n^2 \pi^2}{l^2}} \right) + 2 \frac{\sin 2n\pi l}{\frac{n^3 \pi^3}{l^3}} \right] - 0$

$\cos n\pi = (-1)^n$

$= \frac{1}{l} \left[2l \left(\frac{(-1)^2}{\frac{n^2 \pi^2}{l^2}} \right) + 0 \right]$

$a_n = \frac{1}{l} \left(\frac{4l^3}{n^2 \pi^2} \right)$

$a_n = \frac{4l^2}{n^2 \pi^2}, b_n = 0$

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Mathematics Page

Q Find F. Series for $f(x) = x^2$ in $(-\pi, \pi)$

Hence, deduce the sum of series.

i) $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

ii) $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

sol $f(x) = x^2$
Limit $(-\pi, \pi)$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

period = up limit + lower limit
= $\pi - (-\pi)$
= 2π

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx$$

$\Downarrow$ (or)
 $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$

$f(x) = x^2$
 $f(-x) = (-x)^2 = x^2 \rightarrow$ Even function.
 $\Rightarrow 2 \int_0^{\pi} f(x) dx$
 $\Rightarrow 2 \int_0^{\pi} x^2 dx$
 $\Rightarrow 2 \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^3}{3}$
 $\Rightarrow \frac{2\pi^3}{3}$

$$= \frac{1}{\pi} \cdot \left[\frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{\pi} \left[\frac{\pi^3}{3} - \frac{(-\pi)^3}{3} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi^3}{3} + \frac{\pi^3}{3} \right] = \frac{1}{\pi} \cdot \frac{2\pi^3}{3} = \frac{2\pi^2}{3}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = 0$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx dx$$

The function $f(x)$ is odd.
= 0

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx dx = 0$$

$$= \frac{2}{\pi} \left[x^2 \frac{\sin nx}{n} - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[x^2 \frac{\sin nx}{n} + 2x \frac{\cos nx}{n^2} + 2 \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\sin nx [x^2 + 2] + 2x \cos nx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(0 + 2\pi(-1)^n) - (0 + 0) \right]$$

$$= \frac{2}{\pi} \left[2\pi(-1)^n \right] = 4(-1)^n$$

Fourier series:

$$f(x) = \frac{2}{3} \pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx + 0$$

$$f(x) = \frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} \cos x + \frac{1}{2^2} \cos 2x - \frac{1}{3^2} \cos 3x + \dots \right]$$

For deduction:

$$x = \pi$$

sign are alternate
is alternate
so, we take it.

$$= \frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} \cos \pi + \frac{1}{2^2} \cos 2\pi - \frac{1}{3^2} \cos 3\pi + \frac{1}{4^2} \cos 4\pi - \dots \right]$$

$$= \frac{\pi^2}{3} + 4 \left[+\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \right]$$

$\therefore$ Hence, deduced
Half Range Series:

Find Half range sine series of $f(x) = x(\pi-x)$ in $(0, \pi)$.
Hence, deduce that $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots = \frac{\pi^3}{32}$

Half range - sine series $\rightarrow$ only $\sin$.

Half range - cosine series $\rightarrow$ only $\cos$.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cdot \sin nx \, dx \quad \begin{matrix} \sin \pi = 0 \\ \cos \pi = -1 \end{matrix}$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} (x-x^2) \sin nx \, dx$$

$$= \frac{2}{\pi} \left[(x-x^2) \left(\frac{\cos nx}{n} \right) - (\pi-2x) \left(-\frac{\sin nx}{n^2} \right) + (-2) \left(\frac{\cos nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-(\pi-x^2) \frac{\cos nx}{n} + (\pi-2x) \frac{\sin nx}{n^2} - 2 \frac{\cos nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[0 + 0 - 2 \frac{(-1)^n}{n^3} - \left(-\frac{2}{n^3} \right) \right]$$

$$= \frac{2}{\pi} \left[-2 \frac{(-1)^n}{n^3} + \frac{2}{n^3} \right] = \frac{4}{\pi n^3} \left[\frac{(-1)^n}{1 - (-1)^n} \right]$$

Now, applying in Half range series:

$$f(x) = x(\pi-x) \text{ in } (0, \pi)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{4}{\pi n^3} (1 - (-1)^n) \sin nx \, dx$$

$$= \sum_{n=1}^{\infty} \frac{4}{\pi} \left[\frac{1 - (-1)^n}{n^3} \right] \sin nx \, dx$$

Deduction!

$$f(x) = \frac{4}{\pi} \left[\frac{2}{1^3} \sin x + 0 + \frac{2}{3^3} \sin 3x + 0 + \frac{2}{5^3} \sin 5x + \dots \right]$$

Substitute $x = \pi/2$ for sine.

$$= \frac{8}{\pi} \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \right]$$

for $\sin$
deduction part,
always choose
 $x = \pi/2$

Module-3

~~Fourier Transform:~~
~~Fourier Transform:~~ $F(s) = \int_{-\infty}^{\infty} f(x) e^{isx} dx$
~~Inverse Fourier Transform:~~ $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$

$e^{i\theta} = \cos\theta + i\sin\theta$
 $e^{-i\theta} = \cos\theta - i\sin\theta$

Fourier Transform:-

$$F[f(x)] = F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Inverse - Fourier Transform:

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$$

Parseval's Identity:

If $F(s)$ is the complex Fourier Transform of $f(x)$, then:

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$$

1

S.T: Fourier Transform of $f(x) = \begin{cases} a^2 - x^2, & |x| < a \\ 0, & |x| > a > 0. \end{cases}$

$$is \quad 2 \sqrt{\frac{2}{\pi}} \left(\frac{\sin as - as \cos as}{s^3} \right)$$

2

$$|x| < a \Rightarrow -a < x < a$$

$$\int_{-a}^a (a^2 - x^2) dx$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a (a^2 - x^2) e^{isx} dx$$

$$\rightarrow \text{Let } u = a^2 - x^2$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a (a^2 - x^2) (\cos sx + i \sin sx) dx$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[\int_{-a}^a (a^2 - x^2) \cos sx dx + \int_{-a}^a (a^2 - x^2) i \sin sx dx \right]$$

$\begin{matrix} \text{even} \times \text{even} = \text{even} \\ \text{even} \times \text{odd} = \text{odd} \end{matrix}$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[\int_{-a}^a (a^2 - x^2) \cos sx dx \right]$$

0: odd

$$\rightarrow \text{Let } u = a^2 - x^2$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[(a^2 - x^2) \left(\frac{\sin sx}{s} \right) - (-2x) \left(\frac{-\cos sx}{s^2} \right) + 2 \left(\frac{\sin sx}{s^3} \right) \right]_{-a}^a$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[(a^2 - x^2) \left(\frac{\sin sx}{s} \right) - (-2x) \left(\frac{-\cos sx}{s^2} \right) + 2 \left(\frac{\sin sx}{s^3} \right) \right]_a^{-a}$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[(a^2 - x^2) \left(\frac{\sin sx}{s} \right) + (2x) \left(\frac{-\cos sx}{s^2} \right) + 2 \frac{\sin sx}{s^3} \right]_a^{-a}$$

→ 0 (on applying both upper limit and lower limit)

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[(a^2 - x^2) \left(\frac{\sin sx}{s} \right) - \frac{2x \cos sx}{s^2} + 2 \frac{\sin sx}{s^3} \right]_a^{-a}$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[-\frac{2x \cos sx}{s^2} + \sin sx \left[\frac{a^2 - x^2}{s} + \frac{2}{s^3} \right] \right]_a^{-a}$$

→ DO NOT TAKE COMMON !!!

~~$$f(s) = \frac{1}{\sqrt{2\pi}} \left[-\frac{2a \cos sa}{s^2} + \sin sa \left[0 + \frac{2}{s^3} \right] - \left[\frac{2a \cos sa}{s^2} + \sin sa \left[0 + \frac{2}{s^3} \right] \right] \right]$$~~

~~$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{2a \cos as}{s^2} - \frac{2}{s^3} \sin as - \frac{2a \cos as}{s^2} + \frac{2 \sin as}{s^3} \right]$$~~

~~$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{4a \cos as}{s^2} \right]$$~~

→ on applying both upper and lower limits it becomes zero.

$$f(s) = \frac{1}{\sqrt{2\pi}} \left[0 - \frac{2x \cos sx}{s^2} + 2 \frac{\sin sx}{s^3} \right]_a^{-a}$$

↑ YT: Quantum Academy Madurai

~~$$f(s) = \frac{1}{\sqrt{2\pi}} \left[-2a \frac{\cos sa}{s^2} + \frac{2 \sin sa}{s^3} - \left(+2a \frac{\cos sa}{s^2} - \frac{2 \sin sa}{s^3} \right) \right]$$~~

~~$$= \frac{1}{\sqrt{2\pi}} \left[\frac{-4a \cos sa}{s^2} \right]$$~~

Change the limits at the start

~~$$= \frac{4}{\sqrt{2\pi}} \left[-\frac{a \cos as}{s^2} + \frac{\sin as}{s^3} \right]$$~~

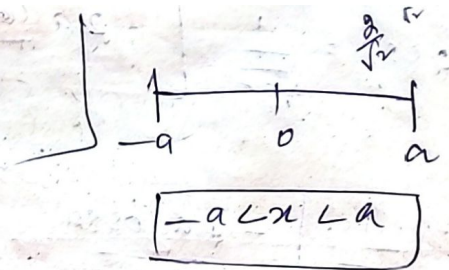
~~$$= \frac{4}{\sqrt{2\pi}} \left[\frac{-as \cos as + \sin as}{s^3} \right] \Rightarrow 2 \sqrt{\frac{2}{\pi}} \left[\frac{\sin as - as \cos as}{s^3} \right]$$~~

∴ Hence, Result is derived.

Note
for constant limits
 $\int_{-a}^a f(x) dx$
 $= 2 \int_0^a f(x) dx$

Find Fourier Transform:

$$f(x) = \begin{cases} x & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$$



Q1

$$f(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cdot e^{isx} dx$$

$$f(s) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a x \cdot e^{isx} dx$$

$$\begin{aligned} e^{i\theta} &= \cos\theta - i\sin\theta \\ e^{-i\theta} &= \cos\theta + i\sin\theta \end{aligned}$$

$$\therefore \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

$$f(s) = \frac{2}{\sqrt{2\pi}} \int_0^a x (\cos sx + i \sin sx) dx$$

$$f(s) = \frac{2}{\sqrt{2\pi}} \left[\int_0^a x \cos sx dx + i \int_0^a x \sin sx dx \right]$$

odd even
odd odd = even
odd

$$f(s) = \frac{2i}{\sqrt{2\pi}} \left[\int_0^a x \sin sx dx \right] \Rightarrow f(s) = \frac{2i}{\sqrt{2\pi}} \left[-x \frac{\cos sx}{s} + \frac{1}{s^2} \sin sx \right]_0^a$$

$$f(s) = \frac{2i}{\sqrt{2\pi}} \left[-a \frac{\cos as}{s} + \frac{\sin as}{s^2} - 0 \right]$$

$$f(s) = \frac{2i}{\sqrt{2\pi}} \left[\frac{-as \cos as + \sin as}{s^2} \right] = \frac{-2i}{s^2 \sqrt{2\pi}} (as \cos as - \sin as)$$

$$f(s) = i \sqrt{\frac{2}{\pi}} \left(\frac{\sin as - as \cos as}{s^2} \right)$$

- Fourier Sine Transform, Cosine transform
- Parseval's identity
- Fourier Integral (sine cosine)

$$\begin{aligned} \frac{2}{\sqrt{2\pi}} &= \frac{\sqrt{2} \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{\pi}} \\ &= \sqrt{\frac{2}{\pi}} \end{aligned}$$

✓

z-transform

$$\textcircled{*} \quad z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

① Find z transform: i) $z[a^n]$, ii) $z[1]$, iii) $z[a]$

$$\textcircled{sol} \quad i) \quad z[a^n] = \sum_{n=0}^{\infty} a^n \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{a^n}{z^n} = \left(\frac{a}{z}\right)^n$$

$$z[a^n] = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n = 1 + \left(\frac{a}{z}\right) + \left(\frac{a}{z}\right)^2 + \dots$$

$$= \left(1 - \frac{a}{z}\right)^{-1} \quad \text{if } \left|\frac{a}{z}\right| < 1$$

$$= \left(\frac{z-a}{z}\right)^{-1} \quad \text{if } \left|\frac{a}{z}\right| < 1$$

$$z[a^n] = \left(\frac{z}{z-a}\right) \quad \text{if } |a| < |z|$$

ii) $z[1]$

$$z[1] = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = \frac{1}{z} = \left(\frac{1}{z}\right)^n$$

$$= 1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \dots \quad (1-x)^{-1} \text{ if } |x| < 1$$

$$= \left(1 - \frac{1}{z}\right)^{-1} \quad \text{if } |z| < 1$$

$$= \left(\frac{z-1}{z}\right)^{-1} \quad \text{if } \left(\frac{1}{z}\right) < 1$$

$$= \left(\frac{z}{z-1}\right) \quad \text{if } 1 < z$$

④ If there is any constant in product
eg: $z[2a]$
we can make it as
 $2 \cdot z[a]$

$$iii) \quad z[a] = \sum_{n=0}^{\infty} a \cdot z^{-n} \Rightarrow \sum_{n=0}^{\infty} \frac{a}{z^n} \quad \text{or} \quad \sum_{n=0}^{\infty} a \left[\frac{1}{z}\right]^n$$

$$z[a] = \sum_{n=0}^{\infty} a \left[\frac{1}{z}\right]^n \Rightarrow \sum_{n=0}^{\infty} a \left[\frac{1}{z}\right]^n$$

$$z[a] = a(1) + \frac{a}{z} + \frac{a}{z^2} + \frac{a}{z^3} + \dots$$

$$z[a] = z[a \cdot 1] = a \cdot z[1]$$

$$\therefore z[1] = \left(\frac{z}{z-1}\right) \text{ from previous question,}$$

$$\therefore z[1] = \frac{z}{z-1}$$

$$z[a] = a \left(\frac{z}{z-1}\right)$$

$\Rightarrow$ Convolution theorem in z-transform & Diff. Equations

$$\text{Convolution theorem: } f(n) * g(n) = \sum_{r=0}^n f(r) g(n-r)$$

Formulas used:

$$z^{-1} \left(\frac{z}{z-a}\right) = a^n; \quad z^{-1} \left(\frac{z}{z+a}\right) = (-a)^n$$

① Find the z -transform of $\frac{z^2}{(z-a)^2}$ by using convolution theorem.

→ Inverse z -transform: find for $\frac{4z^2 - 2z}{(z-1)(z-2)^2}$ using partial fraction Method.

Q) Step 1: let $F(z) = \frac{4z^2 - 2z}{(z-1)(z-2)^2}$... $\div$ by z in numerator

$$\frac{F(z)}{z} = \frac{4z - 2}{(z-1)(z-2)^2}$$

Now, partial fraction

$$\frac{4z-2}{(z-1)(z-2)^2} = \frac{A}{(z-1)} + \frac{B}{(z-2)} + \frac{C}{(z-2)^2} \quad \text{--- (1)}$$

$$\frac{4z-2}{(z-1)(z-2)^2} = \frac{A(z-1)(z-2)^2}{(z-1)(z-2)^2} + \frac{B(z-2)(z-1)}{(z-1)(z-2)^2}$$

$$\frac{4z-2}{(z-1)(z-2)^2} = \frac{A(z-1)(z-2)^2 + B(z-1)(z-2) + C(z-1)}{(z-1)(z-2)^2}$$

YT
Sk Maths
tutor

$$A(z-2)^2 + B(z-1)(z-2) + C(z-1) = 4z-2$$

Put $z=1$

$$A(-1)^2 + 0 + 0 = 4-2$$

$$\boxed{A=2}$$

Must check all formulas of partial fraction

Put $z=2$

$$A(0) + 0 + C(1) = 8-2$$

$$\boxed{C=6}$$

Put $z=0$

$$A(-2)^2 + B(-1)(-2) + C(-1) = -2$$

$$4A + 2B - C = -2$$

→ Substituting A and C , we get

$$4(2) + 2B - 6 = -2$$

$$2B + 8 - 6 = -2$$

$$2B + 2 = -2$$

$$2B = -4$$

$$\boxed{B=-2}$$

Now substituting A, B, C in Eq. (1)

$$\frac{F(z)}{z} = \frac{2}{z-1} + \frac{-2}{z-2} + \frac{6}{(z-2)^2}$$

$$= 2\left(\frac{z}{z-1}\right) - 2\left(\frac{z}{z-2}\right) + 6\left(\frac{z}{(z-2)^2}\right)$$

Formulas:

$$z^{-1} \left[\frac{z}{z-a} \right] = a^n$$

$$z^{-1} \left[\frac{z}{z+a} \right] = (-a)^n$$

$$z^{-1} \left[\frac{z}{(z-a)^2} \right] = na^{n-1}$$

$$= 2(1)^n - 2(2)^n + 6(n(2)^{n-1})$$

$$= 2(1)^n - 2(2)$$

① Inverse Z transform of $\frac{z^2}{(z-a)^2}$ by using convolution.

$$\begin{aligned} z^{-1} \left[\frac{z^2}{(z-a)^2} \right] &= z^{-1} \left[\left(\frac{z}{z-a} \right) \cdot \left(\frac{z}{z-a} \right) \right] \\ &= z^{-1} \left(\frac{z}{z-a} \right) * z^{-1} \left(\frac{z}{z-a} \right) \\ &= a^n * a^n \\ &= f(n) * g(n); f(n) = a^n \text{ \& } g(n) = a^n. \end{aligned}$$

$$\therefore f(n) * g(n) = \sum_{r=0}^n f(r) \cdot g(n-r).$$

$$= \sum_{r=0}^n a^r \cdot a^{n-r}.$$

$$= \sum_{r=0}^n a^r \cdot a^n \cdot a^{-r}$$

$$= \sum_{r=0}^n a^n \cdot \frac{1}{a^r}$$

$$= \sum_{r=0}^n a^n \cdot [1 + 1 + 1 + \dots + 1]$$

$$= a^n \sum_{r=0}^n 1 = a^n (n+1)$$

Module - 5

① Solve the difference equation $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$

with $y_0 = 0, y_1 = 1$ using Z transform.

② Take Z-transform on B.S

$$z[y_{n+2} + 4y_{n+1} + 3y_n] = z[3^n]$$

$$z(y_{n+2}) + 4z(y_{n+1}) + 3z(y_n) = \frac{z}{z-3}$$

③ Important:

$$z[y(n+2)] = z^2 y(n) - z^2 y(0) - z y(1)$$

$$z[y(n+1)] = z y(n) - z y(0)$$

$$z[y(n)] = y(n)$$

$$\begin{aligned} z[a^n] &= \frac{z}{z-a} \end{aligned}$$

$$[z^2 y(n) - z^2 y(0) - z y(1)] + 4[z y(n) - z y(0)] + 3y(n) = \frac{z}{z-3}$$

Given that: $y_0 = 0, y_1 = 1$.

④ Answer wrong

$$z^2 y(n) - 0 - z(1) + 4(y(n) - 0) + 3y(n) = \frac{z}{z-3}$$

$$z^2 y(n) - z + 4y(n) + 3y(n) = \frac{z}{z-3}$$

$$z^2 y(n) + 7y(n) - z = \frac{z}{z-3}$$

$$y(n) [z^2 + 7] = \frac{z}{z-3} + z$$

$$y(n) = \frac{z^2 - 2z}{(z-3)(z^2+7)}$$

$$\frac{z^2 + 7z - 3z}{z-3} = \frac{z^2 + 4z}{z-3}$$

pt :

Application of z-transform to solve linear Difference Equ.

Procedure:

① $z\{y_n\} = F(z)$

② Apply z-transform on both sides using:

③ $z[y_n] = F(z)$
 $z[y_{n+1}] = z[F(z) - y_0]$
 $z[y_{n+2}] = z^2[F(z) - y_0 - \frac{y_1}{z}]$
 $z[y_{n+3}] = z^3[F(z) - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2}]$

④ Using given conditions, simplify.

⑤ Express $F(z)$ in terms of z transform.

⑥ Take Inverse z-transform on both sides.

⑦ Use Partial fraction & simplify

⑧ Isolate y_n as a function of n which is the solution.

⑨ Solve the Differential Equation:

$$y_{n+2} + 2y_{n+1} + y_n = n ; y_0 = y_1 = 0$$

Sol

$$y_{n+2} + 2y_{n+1} + y_n = n$$

Applying z-transform on B-S

$$z[y_{n+2} + 2y_{n+1} + y_n] = z[n]$$

$$z(y_{n+2}) + 2z(y_{n+1}) + z(y_n) = z(n)$$

$$[F(z) - y_0 - \frac{y_1}{z}] + 2[F(z) - y_0] + zF(z) = \frac{z}{(z-1)^2}$$

$$F(z) - y_0 - \frac{y_1}{z} + 2F(z) - 2y_0 + zF(z) = \frac{z}{(z-1)^2}$$

$$3F(z) + zF(z) - 3y_0 - \frac{y_1}{z} = \frac{z}{(z-1)^2}$$

$$3F(z) + zF(z) = \frac{z}{(z-1)^2} + \frac{3y_0 + y_1}{z}$$

∴ must use - residue method.

Full module same concepts.

⑩ must learn